

Name: _____

Class: _____

Date: _____

Unit 3 Review Packet

Algorithms & Programming



How to use this packet

Match the vocabulary, then work the mixed review questions. These are the exact words and question types you'll see on the unit test and the AP exam.

Vocab Match 1

- _____ 1. **Variable**
- _____ 2. **Assignment**
- _____ 3. **Data type**
- _____ 4. **Boolean**
- _____ 5. **String**
- _____ 6. **List**
- _____ 7. **Element**
- _____ 8. **Index**
- _____ 9. **Data abstraction**
- _____ 10. **Empty list**

- A.** managing complexity by naming a collection of data and hiding the details
- B.** an individual value in a list
- C.** a list with no elements, written []
- D.** storing a value in a variable; replaces whatever was there before
- E.** the kind of value (number, string, Boolean, list); determines what operations make sense
- F.** an ordered sequence of elements stored under one name
- G.** a named storage location for a value that can change
- H.** a value that is either true or false
- I.** an ordered sequence of characters
- J.** a common way to reference elements by their position in a list

Vocab Match 2

- _____ 1. **Algorithm**
- _____ 2. **Sequencing**
- _____ 3. **Selection**
- _____ 4. **Iteration**
- _____ 5. **Expression**
- _____ 6. **String**
- _____ 7. **Concatenation**
- _____ 8. **Substring**
- _____ 9. **Empty string**
- _____ 10. **Length**

- A.** repeating steps
- B.** steps run one after another in order
- C.** an ordered sequence of characters
- D.** joining strings together end-to-end
- E.** the number of characters in a string
- F.** a part of an existing string
- G.** choosing which steps to run based on a condition
- H.** a string with no characters, written ""
- I.** a value, variable, operator, or procedure call that is evaluated to produce a single value
- J.** a finite set of instructions that accomplish a specific task

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Vocabulary (continued)

Vocab Match 3

- _____ 1. **Boolean value**
- _____ 2. **Boolean expression**
- _____ 3. **Relational operator**
- _____ 4. **Logical operator**
- _____ 5. **Equivalent expressions**
- _____ 6. **Selection**
- _____ 7. **Conditional statement**
- _____ 8. **Block**
- _____ 9. **Flow of control**
- _____ 10. **Nested conditional**

- A. two expressions that produce the same result for every input
- B. changes the sequential flow of control by running different statements based on a Boolean expression
- C. NOT, AND, OR — combines or reverses Boolean values
- D. the order in which statements are executed
- E. an expression that evaluates to true or false
- F. a group of statements inside braces { }
- G. a conditional statement placed inside another conditional
- H. determines which parts of an algorithm run based on a condition being true or false
- I. compares two values ($=$, $\neq$, $>$, $<$, $\geq$, $\leq$) and produces a Boolean
- J. a value that is either true or false

Vocab Match 4

- _____ 1. **Branch**
- _____ 2. **Compound condition**
- _____ 3. **Decision tree**
- _____ 4. **Equivalent code**
- _____ 5. **Iteration**
- _____ 6. **Infinite loop**
- _____ 7. **Traversal**
- _____ 8. **Counter**
- _____ 9. **Accumulator**
- _____ 10. **Algorithm**

- A. a repeating portion of an algorithm; repeats a set number of times or until a condition is met
- B. a variable that builds up a total as the loop runs
- C. accessing each element of a list, one at a time
- D. different code that produces the same output for all inputs
- E. a finite set of instructions that accomplish a task
- F. a loop whose ending condition never becomes true
- G. a variable that tracks how many times something happens
- H. a diagram that shows each choice as a fork in the path
- I. a condition that combines Booleans with AND / OR / NOT
- J. one possible path through a conditional

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Vocabulary (continued)

Vocab Match 5

- _____ 1. Pseudocode
- _____ 2. Flowchart
- _____ 3. Equivalent algorithms
- _____ 4. Building block
- _____ 5. Traversal
- _____ 6. Linear search
- _____ 7. APPEND
- _____ 8. INSERT
- _____ 9. REMOVE
- _____ 10. Binary search

- A. repeatedly checks the middle and eliminates half of a sorted list
- B. visiting the elements of a list one at a time
- C. an existing, correct algorithm reused to create a new one
- D. adds a value at a specific index and shifts the rest right
- E. a diagram of an algorithm using shapes and arrows
- F. checks each element in order; works on any list
- G. deletes the element at an index; length decreases by 1
- H. algorithms that give the same output for every input
- I. a plain-language description of code that isn't tied to one programming language
- J. adds a value to the end of a list

Vocab Match 6

- _____ 1. Linear search
- _____ 2. Sorted
- _____ 3. Worst case
- _____ 4. Efficiency
- _____ 5. Procedure
- _____ 6. Parameter
- _____ 7. Argument
- _____ 8. Return value
- _____ 9. Procedure call
- _____ 10. Procedural abstraction

- A. the actual value passed in when calling
- B. the value a procedure sends back to where it was called
- C. a named group of instructions that can be called
- D. checks elements one by one in order
- E. how much time / resources an algorithm uses as input grows
- F. using a procedure by its name and behavior, hiding how it works
- G. arranged in order (numeric or alphabetical)
- H. a placeholder variable in the procedure definition
- I. a statement that runs a procedure, e.g. area(3, 7)
- J. the input that requires the most steps

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Vocabulary (continued)

Vocab Match 7

- _____ 1. **Modularity**
- _____ 2. **Generalization**
- _____ 3. **Subproblem**
- _____ 4. **Duplicate code**
- _____ 5. **Library**
- _____ 6. **API**
- _____ 7. **Documentation**
- _____ 8. **Procedural abstraction**
- _____ 9. **Import**
- _____ 10. **Random number generator**

- A. produces values that can't be predicted
- B. breaking a program into separate parts / modules
- C. a specification of how a library's procedures behave and are used
- D. using parameters so one procedure works for many inputs
- E. a smaller piece of a larger problem
- F. explains each procedure's name, parameters, and return value
- G. copied code that should be replaced by a procedure
- H. a collection of procedures ready to be used in new programs
- I. you can use a procedure without knowing how it works
- J. a statement that makes a library's procedures available in your program

Vocab Match 8

- _____ 1. **Inclusive**
- _____ 2. **Probability**
- _____ 3. **Outcome**
- _____ 4. **Pseudorandom**
- _____ 5. **Simulation**
- _____ 6. **Hypothesis**
- _____ 7. **Variability**
- _____ 8. **Assumption**
- _____ 9. **Abstraction**
- _____ 10. **Problem**

- A. the range includes both endpoints
- B. how likely an outcome is, from 0 to 1
- C. one possible result of a random event
- D. a program that imitates a real-world process or system
- E. a general description of a task that may or may not be solved with an algorithm
- F. computer-generated numbers that look random but come from an algorithm
- G. natural randomness or variation in real-world events
- H. something the model treats as true to simplify reality
- I. a prediction that can be tested
- J. reducing complexity by hiding details and keeping what matters

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Vocabulary (continued)

Vocab Match 9

- | | |
|--------------------------------------|---|
| _____ 1. Instance | A. deciding whether any program will finish running or run forever on a given input |
| _____ 2. Decision problem | B. an algorithm can solve every instance correctly |
| _____ 3. Optimization problem | C. a problem with a yes/no answer |
| _____ 4. Heuristic | D. a problem with a specific input |
| _____ 5. Decision problem | E. a problem with a yes or no answer |
| _____ 6. Decidable | F. no algorithm can solve every instance correctly |
| _____ 7. Undecidable | G. an approach that finds a solution that is not guaranteed to be optimal, used when the exact solution is impractical |
| _____ 8. Halting problem | H. the problem with one specific input |
| _____ 9. Instance | I. a problem that asks for the best solution |

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1. A sorted list grows from 1,000 items to 2,000 items. How does the worst-case number of binary search comparisons change?

- A. It doubles
- B. It increases by about 1
- C. It increases by 1,000
- D. It stays exactly the same

2. Which statement is true?

- A. Some instances of an undecidable problem may be solvable by an algorithm
- B. An undecidable problem can be solved with a fast enough computer
- C. Every decision problem is decidable
- D. Undecidable problems have no correct answers

3. What is the value of x?

```
PROCEDURE double(n)
{ RETURN (n * 2) }
x ← double(double(3))
```

- A. 12
- B. 9
- C. 6
- D. 36

4. What is displayed?

```
a ← 10
b ← a
a ← 20
DISPLAY(b)
```

- A. b
- B. 20
- C. 30
- D. 10

5. Why might a programmer add a parameter to a procedure?

- A. To make the procedure run faster
- B. To prevent the procedure from returning a value
- C. So the procedure no longer needs to be called
- D. So one procedure can be reused with different input values

6. What is displayed?

```
x ← 4
IF (x > 5)
{ DISPLAY("A") }
ELSE
{ DISPLAY("B") }
```

- A. A
- B. Nothing
- C. A B
- D. B

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7. What is displayed?

```
result ← 0
i ← 1
REPEAT UNTIL (i > 10)
{
  IF (i MOD 2 = 0)
  { result ← result + i }
  i ← i + 1
}
DISPLAY(result)
```

- A. 25
- B. 20
- C. 55
- D. 30

8. This code is meant to find the minimum of nums.

```
min ← 0
FOR EACH n IN nums
{
  IF (n < min) { min ← n }
}
```

For which list does it work correctly?

- A. [5, 3, 9]
- B. [-2, -8, 4]
- C. [1, 2, 3]
- D. [7]

9. What is displayed?

```
a ← 7
b ← 3
a ← a MOD b + a
b ← a MOD (b + 2)
a ← (a + b) MOD 4
DISPLAY(a)
DISPLAY(b)
```

- A. 0 1
- B. 2 3
- C. 3 3
- D. 3 8

10. A store gives a discount if a customer is a member AND spent at least \$50, OR if it's the customer's birthday. Which expression is correct?

- A. isMember AND (spent ≥ 50 OR isBirthday)
- B. (isMember AND spent ≥ 50) OR isBirthday
- C. isMember OR spent ≥ 50 OR isBirthday
- D. (isMember OR spent ≥ 50) AND isBirthday

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11. Which TWO code segments correctly swap the values of `a` and `b`? (Select two.)

I. `temp ← a`
`a ← b`
`b ← temp`

II. `a ← b`
`b ← a`

III. `a ← a + b`
`b ← a - b`
`a ← a - b`

IV. `temp ← a`
`a ← b`
`b ← a`

- A. I
- B. II
- C. III
- D. IV

12. Algorithm I adds all the numbers and then divides by how many there are. Algorithm II divides each number by the count and then adds the results. Which is true?

- A. Only I computes the average
- B. Both compute the average
- C. Only II computes the average
- D. Neither computes the average

13. Which loop is an **infinite loop**?

- A. `i ← 1, REPEAT UNTIL (i = 10) { i ← i + 2 }`
- B. `i ← 1, REPEAT UNTIL (i > 10) { i ← i + 2 }`
- C. `i ← 10, REPEAT UNTIL (i < 0) { i ← i - 1 }`
- D. `i ← 0, REPEAT UNTIL (i ≥ 5) { i ← i + 1 }`

14. What are the values after this code runs?

```
p ← 10
q ← 15
r ← 20
p ← q
q ← r
r ← p
```

- A. `p = 15, q = 20, r = 10`
- B. `p = 20, q = 15, r = 10`
- C. `p = 10, q = 15, r = 20`
- D. `p = 15, q = 20, r = 15`

15. Using the AP reference sheet, what does `a` contain after this code?

```
a ← [2, 4, 6, 8]
b ← a
b[2] ← 10
```

- A. `[2, 4, 6, 8]`
- B. `[2, 10, 6, 8]`
- C. `[10, 4, 6, 8]`
- D. `[2, 4, 10, 8]`

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16. What is result after this code?

```
list1 ← [10, 20, 30, 40]
result ← []
FOR EACH x IN list1
{
  IF (x MOD 20 = 0)
  { APPEND(result, x / 10) }
  ELSE
  { APPEND(result, x) }
}
```

- A. [1, 2, 3, 4]
- B. [2, 4]
- C. [10, 2, 30, 4]
- D. [10, 20, 30, 40]

18. What is the value of $4 + 8 \text{ MOD } 3 * 2$?

- A. 6
- B. 8
- C. 12
- D. 2

20. Which expression evaluates to the ones digit of a positive integer n ?

- A. $n / 10$
- B. $10 \text{ MOD } n$
- C. $n \text{ MOD } 10$
- D. $n * 10$

17. A program calls `calculateTax(price)` in 20 places. A developer rewrites the inside of `calculateTax` to run faster, and it returns the same results as before. What else must change?

- A. All 20 calls must be rewritten
- B. Every procedure in the program must be rewritten
- C. The parameter must be renamed
- D. Nothing — the calls still work the same way

19. A robot faces **up**. Where does it end up?

```
REPEAT 3 TIMES
{
  MOVE_FORWARD()
  ROTATE_RIGHT()
}
```

- A. Back at the start, facing up
- B. One square right of the start, facing left
- C. One square above the start, facing right
- D. Three squares above the start, facing up

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21. What is displayed?

```
PROCEDURE f(x)
{
  x ← x + 5
  RETURN (x * 2)
}
a ← 3
b ← f(a)
DISPLAY(a)
DISPLAY(b)
```

- A. 3 16
- B. 8 16
- C. 3 6
- D. 8 8

22. Which TWO segments set `max` to the larger of `a` and `b` for ALL values, including when `a = b`? (*Select two.*)

```
I. IF (a > b) { max ← a }
   ELSE { max ← b }
II. max ← a
   IF (b > max) { max ← b }
III. IF (a > b) { max ← a }
     IF (b > a) { max ← b }
IV. max ← b
    IF (a < b) { max ← a }
```

- A. I
- B. II
- C. III
- D. IV

23. A flu simulation assumes every student has the same number of contacts per day. What is the most likely consequence?

- A. The results may not reflect real spread, because some students have many more contacts
- B. The results will exactly match real life
- C. The simulation will run forever
- D. The simulation will not need random numbers

24. A spinner is red 50%, blue 25%, green 25%. Which code simulates it correctly?

```
I. n ← RANDOM(1, 4)
   IF (n ≤ 2) { c ← "red" }
   ELSE { IF (n = 3) { c ← "blue" }
          ELSE { c ← "green" } } }
II. IF (RANDOM(1, 4) ≤ 2) { c ← "red" }
     ELSE { IF (RANDOM(1, 4) = 3) { c ← "blue" }
            ELSE { c ← "green" } } }
```

- A. I only
- B. II only
- C. Both I and II
- D. Neither

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25. What is displayed?

```
x ← 12
IF (x > 10)
{
  IF (x > 20) { DISPLAY("A") }
  ELSE { DISPLAY("B") }
}
ELSE { DISPLAY("C") }
```

- A. A
- B. B C
- C. C
- D. B

27. Which expression is equivalent to $\text{NOT } (x > 10 \text{ OR } y < 5)$?

- A. $(x < 10) \text{ OR } (y > 5)$
- B. $(x < 10) \text{ AND } (y > 5)$
- C. $(x \leq 10) \text{ OR } (y \geq 5)$
- D. $(x \leq 10) \text{ AND } (y \geq 5)$

29. What is displayed?

```
PROCEDURE firstOver(list, limit)
{
  FOR EACH item IN list
  {
    IF (item > limit) { RETURN (item) }
  }
  RETURN (-1)
}
DISPLAY(firstOver([4, 9, 2, 12], 5))
DISPLAY(firstOver([1, 2], 5))
```

- A. -1 -1
- B. 12 -1
- C. 9 12
- D. 9 -1

26. What does it mean for a problem to be **undecidable**?

- A. It takes a long time to solve
- B. It can only be solved by a heuristic
- C. It has no inputs
- D. No algorithm can always give a correct yes-or-no answer for all inputs

28. Which statement about simulations is true?

- A. They can test hypotheses without real-world constraints
- B. They always produce exact predictions
- C. They cannot use random numbers
- D. They must include every real-world detail

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30. What is the probability that this code displays 3?

```
count ← 0
REPEAT 3 TIMES
{
  IF (RANDOM(1, 2) = 1)
  { count ← count + 1 }
}
DISPLAY(count)
```

- A. 1/2
- B. 1/8
- C. 1/6
- D. 1/3

32. Which value can `RANDOM(3, 8)` NOT produce?

- A. 3
- B. 9
- C. 8
- D. 5

34. What is the value of `n` after this code runs when `n` starts at 7?

```
IF (n MOD 2 = 0)
{ n ← n / 2 }
ELSE
{ n ← 3 * n + 1 }
```

- A. 3.5
- B. 7
- C. 21
- D. 22

31. Which expression ALWAYS gives the last element of a non-empty list `nums`?

- A. `nums[LENGTH(nums) - 1]`
- B. `nums[LENGTH(nums)]`
- C. `nums[0]`
- D. `nums[LENGTH(nums) + 1]`

33. Which TWO statements are true? (*Select two.*)

- A. Using library code requires understanding exactly how each procedure is implemented.
- B. Libraries can reduce development time.
- C. API documentation describes how to use a library's procedures.
- D. Libraries can only contain procedures written by the same programmer.

35. What is an API?

- A. A type of computer virus
- B. A specification for how procedures in a library behave and can be used
- C. A programming language
- D. A list of every bug in a program

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36. What is a benefit of storing 30 test scores in a list instead of 30 separate variables?

- A. The scores can be processed with a loop, and the code works if the number of scores changes
- B. Lists use no memory
- C. Lists can only hold numbers, which prevents errors
- D. Each score gets its own name

38. How many characters are in the string "AP CSP 2027" ?

- A. 11
- B. 10
- C. 9
- D. 8

40. Which TWO statements are true? (Select two.)

- A. Binary search can be used on any list.
- B. Linear search can be used on an unsorted list.
- C. For a large sorted list, binary search usually needs fewer comparisons than linear search.
- D. Binary search always finds a value faster than linear search.

42. What is displayed?

```
a ← 5
b ← 10
IF (a > b) { a ← b }
ELSE { b ← a }
IF (a = b) { a ← a + b }
DISPLAY(a)
DISPLAY(b)
```

- A. 10 5
- B. 5 10
- C. 10 10
- D. 15 10

37. A programmer wants to use a library procedure `sortList`. Which information is MOST necessary?

- A. The algorithm `sortList` uses internally
- B. The name of the person who wrote it
- C. What parameters it takes and what it returns or does
- D. How many lines of code it contains

39. What does `concat("data", "base")` return?

- A. "data base"
- B. "basedata"
- C. "database"
- D. "data" "base"

41. Which expression best simulates the sum of rolling two fair six-sided dice?

- A. `RANDOM(2, 12)`
- B. `RANDOM(1, 6) + RANDOM(1, 6)`
- C. `2 * RANDOM(1, 6)`
- D. `RANDOM(1, 12)`

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43. Which TWO statements are true? (*Select two.*)

- A. Every decision problem is decidable.
- B. An algorithm can determine whether certain specific programs halt.
- C. A problem that takes exponential time is undecidable.
- D. No algorithm can solve all instances of an undecidable problem.

44. A delivery company must find the shortest route through 200 stops. Checking every route would take longer than the age of the universe. What is most appropriate?

- A. Buy a faster computer and check every route
- B. Choose a random route
- C. Give up; the problem is undecidable
- D. Use a heuristic that quickly finds a good, though not necessarily optimal, route

45. Which is a benefit of building a new algorithm from existing, correct algorithms?

- A. It guarantees the new program has no errors
- B. It makes the program run in constant time
- C. It reduces development and testing time
- D. It removes the need for documentation

46. `substring(str, start, length)` returns *length* characters starting at position *start* (first position is 1). What does

`substring("ALGORITHM", 3, 4)` return?

- A. "ORIT"
- B. "LGOR"
- C. "GORI"
- D. "GOR"

47. This code is meant to swap *x* and *y*:

```
x ← y
y ← x
```

What does it actually do?

- A. Swaps *x* and *y* correctly
- B. Both variables end with the original value of *x*
- C. Both variables end with the original value of *y*
- D. Causes an error

48. Which expression is true exactly when *num* is between 1 and 10, inclusive?

- A. $(\text{num} \geq 1) \text{ OR } (\text{num} \leq 10)$
- B. $\text{NOT } (\text{num} = 1)$
- C. $(\text{num} > 1) \text{ AND } (\text{num} < 10)$
- D. $(\text{num} \geq 1) \text{ AND } (\text{num} \leq 10)$

49. This code should compute the product of all numbers in *nums*, but always displays 0. Which change fixes it?

```
product ← 0
FOR EACH n IN nums
{ product ← product * n }
DISPLAY(product)
```

- A. Change `product * n` to `product + n`
- B. Move `DISPLAY` inside the loop
- C. Initialize `product` to 1 instead of 0
- D. Change `FOR EACH` to `REPEAT UNTIL`

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50. Which segments display "big" when $x > 10$ and "small" otherwise, for ALL integers x ?

- I. IF ($x > 10$) { DISPLAY("big") }
ELSE { DISPLAY("small") }
- II. IF ($x > 10$) { DISPLAY("big") }
IF ($x < 10$) { DISPLAY("small") }
- III. IF ($x \leq 10$) { DISPLAY("small") }
ELSE { DISPLAY("big") }

- A. I only
- B. II only
- C. I and III only
- D. I, II, and III

52. Given PROCEDURE area(w , h) and the call area(3, 7), what are the arguments?

- A. w and h
- B. 3 and 7
- C. area
- D. 21

54. What must be true about a list before binary search can be used?

- A. It must contain only numbers
- B. It must have an even number of elements
- C. It must be sorted
- D. It must have fewer than 1,000 elements

51. What does $(5 > 3)$ AND $(2 > 4)$ evaluate to?

- A. true
- B. Error
- C. 5
- D. false

53. x starts at 1. What is x after this code?

```
REPEAT 5 TIMES  
{  $x \leftarrow x + 2$  }
```

- A. 10
- B. 11
- C. 7
- D. 12

55. Which TWO changes would most likely make a traffic simulation MORE accurate? (Select two.)

- A. Include rush-hour variation in the number of cars
- B. Assume every driver drives exactly the speed limit
- C. Use real traffic counts to set how often cars arrive
- D. Remove all randomness so every run is identical

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56. For which TWO inputs is "Go" displayed?
(Select two.)

```
IF (NOT (light = "red"))  
{  
  IF (light = "green" OR clear)  
  { DISPLAY("Go") }  
}
```

- A. light = "red", clear = true
- B. light = "green", clear = false
- C. light = "yellow", clear = true
- D. light = "yellow", clear = false

57. What does this code do?

```
count ← 0  
FOR EACH w IN words  
{  
  IF (w = "yes")  
  { count ← count + 1 }  
}
```

- A. Counts how many elements equal "yes"
- B. Removes every "yes"
- C. Finds the index of the first "yes"
- D. Counts the words in the list

58. An algorithm takes 4 steps when $n = 2$, 9 steps when $n = 3$, and 16 steps when $n = 4$. Which best describes it?

- A. n steps — reasonable time
- B. 2^n steps — unreasonable time
- C. $n!$ steps — unreasonable time
- D. n^2 steps — reasonable time

59. Which is a main benefit of using libraries?

- A. Programs never contain errors
- B. They make programs use no memory
- C. They simplify creating complex programs
- D. They remove the need for testing

60. Which best explains why the Halting Problem is undecidable?

- A. It has been proven that no algorithm can correctly decide halting for all program-input pairs
- B. Computers aren't fast enough yet
- C. Programs are too long to read
- D. It requires exponential time

61. What does list contain after this code?

```
list ← [1, 2, 3]  
APPEND(list, 4)  
INSERT(list, 1, 0)
```

- A. [1, 0, 2, 3, 4]
- B. [1, 2, 3, 4, 0]
- C. [0, 2, 3, 4]
- D. [0, 1, 2, 3, 4]

Unit 3 Review Packet

Mixed Review

Mixed Review

62. Which best describes procedural abstraction?

- A. Writing every step of a program in one long block
- B. Removing all parameters from procedures
- C. Using a procedure knowing what it does without knowing how it does it
- D. Converting a program to binary

63. About how many elements does binary search examine, in the worst case, in a sorted list of 500 items?

- A. 50
- B. 9
- C. 250
- D. 500

64. Which is equivalent to the code below for all values?

```
IF (a > 0)
{
  IF (b > 0) { DISPLAY("yes") }
}
```

- A. IF (a > 0 OR b > 0) { DISPLAY("yes") }
- B. IF (b > 0) { DISPLAY("yes") }
- C. IF (NOT (a > 0)) { DISPLAY("yes") }
- D. IF (a > 0 AND b > 0) { DISPLAY("yes") }

65. Which is a **decision** problem?

- A. What is the shortest path from A to B?
- B. Is 97 a prime number?
- C. What is the largest value in a list?
- D. What is the best schedule for 30 classes?

66. `myList ← [7, 3, 9]`. What is `myList[1]`?

- A. 3
- B. 7
- C. 9
- D. Error

67. Which statement is true about this code?

```
cost ← 5
IF (weight > 10)
{ cost ← 20 }
ELSE
{
  IF (isMember) { cost ← 0 }
}
```

- A. A non-member with weight 4 pays 5
- B. A member with weight 12 pays 0
- C. A member with weight 10 pays 20
- D. Every member pays 0

Name: _____

Class: _____

Date: _____

Unit 3 Review Packet

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68. Which is the BEST use of a simulation?

- A. Counting the students in a classroom
- B. Sorting a list of names
- C. Looking up a phone number
- D. Testing how a new car design performs in a crash

69. `len(str)` returns the number of characters in `str`. What is displayed?

```
word ← "STAR"
result ← ""
i ← len(word)
REPEAT UNTIL (i < 1)
{
    result ← concat(result,
    substring(word, i, 1))
    i ← i - 1
}
DISPLAY(result)
```

- A. STAR
- B. RATS
- C. SSSS
- D. Error

70. An algorithm's number of steps grows as 2^n . Which best describes it?

- A. Constant time
- B. Linear time
- C. Unreasonable time
- D. Reasonable time

71. What is the value of `13 MOD 4`?

- A. 3
- B. 1
- C. 3.25
- D. 4

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Unit 3 Review Packet

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72. Which procedure correctly returns how many times `target` appears in `list`?

```
I.  count ← 0
    FOR EACH x IN list
      { IF (x = target) { count ← count +
1 } }
    RETURN (count)
II. FOR EACH x IN list
    { count ← 0
      IF (x = target) { count ← count +
1 } }
    RETURN (count)
III. count ← 0
     FOR EACH x IN list
       { IF (x = target) { RETURN (count +
1) } }
     RETURN (count)
```

- A. I only
- B. II only
- C. III only
- D. I and III

Name: _____

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Date: _____

Unit 3 Review Packet

Challenge Questions

Challenge 3.1

Write code that **rotates** three variables: a gets b's value, b gets c's value, and c gets a's original value. Use one temporary variable and as few statements as possible. Then trace it with $a = 1$, $b = 2$, $c = 3$.

Challenge 3.2

A phone stores contacts in two lists, `names` and `numbers`, where `names[i]` goes with `numbers[i]`. Explain what goes wrong if a name is removed from `names` but not from `numbers`. How would storing each contact as one combined record help?

Challenge 3.3

(a) Write a Boolean expression that is true when year `y` is a multiple of 4. (b) A game cycles through a list of 5 players. Write an assignment that moves `turn` from the current player to the next, wrapping from player 5 back to player 1.

Challenge 3.4

Write ONE expression using `concat` and `substring` that builds a school email from variables `first` and `last`: first initial + full last name + `@school.org`. Example: `first = "Jamal"`, `last = "Reed"` → `"JReed@school.org"`.
